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Subject: Multi-Variate Analysis

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Specialization: Business Analytics

Assignment-II (imports_85)

I have assigned with the “imports_85” dataset to work on and impute the missing values in it and prepare a Regression Model to Predict the Car Price.

Here I'm describing the whole step by step working procedure.

I have attached the dataset in R and stored the name of the dataset as

```
b1= imports_85
```

Before Proceeding we need to install some libraries to work on:

```
library(mice)
```

```
library(VIM)
```

```
library(Hmisc)
```

Step 1: Checking missing values:

```
R Code: is.na(b1)
```

```
which(is.na(b1))
```

The output will show the positions of the missing values column wise. If missing value present, then 'TRUE' else 'FALSE'.

Step 2: Summary of the missing values of the Dataset:

```
R Code: summary(is.na(b1))
```

```
sum(is.na(b1))
```

Output: The output result will show the missing Values in the dataset for each variable and column and also the total number of missing values present in the dataset.

Total 59 missing values are present in the dataset.

```

> #Checking the Summary of the Missing Values
> summary(is.na(b1))
symboling      normalized-losses      make      fuel-type      aspiration
Mode :logical  Mode :logical  Mode :logical  Mode :logical  Mode :logical
FALSE:205      FALSE:164      FALSE:205      FALSE:205      FALSE:205
               TRUE :41
num-of-doors    body-style      drive-wheels    engine-location  wheel-base
Mode :logical  Mode :logical  Mode :logical  Mode :logical  Mode :logical
FALSE:203      FALSE:205      FALSE:205      FALSE:205      FALSE:205
               TRUE :2
  length      width      height      curb-weight      engine-type
Mode :logical  Mode :logical  Mode :logical  Mode :logical  Mode :logical
FALSE:205      FALSE:205      FALSE:205      FALSE:205      FALSE:205

num-of-cylinders engine-size      fuel-system      bore      stroke
Mode :logical    Mode :logical  Mode :logical  Mode :logical  Mode :logical
FALSE:205        FALSE:205      FALSE:205      FALSE:201      FALSE:201
               TRUE :4         TRUE :4
compression-ratio horsepower      peak-rpm      city-mpg      highway-mpg
Mode :logical    Mode :logical  Mode :logical  Mode :logical  Mode :logical
FALSE:205        FALSE:203      FALSE:203      FALSE:205      FALSE:205
               TRUE :2         TRUE :2

  price
Mode :logical
FALSE:201
TRUE :4
> sum(is.na(b1))
[1] 59

```

Step 3: Checking Summary of the dataset:

R Code: summary(b1)

Output: The Output will show the summary of the whole dataset.

```

> t(summary(b1))
      symboling      normalized-losses      make      fuel-type      aspiration
Min.   : -2.0000   Min.   : 65          Length:205   Length:205   Length:205
1st Qu.: 0.0000   1st Qu.: 94          Class :character  Mode :character
Median : 1.0000   Median :115          Mode :character  Mode :character
Mean   : 0.8341   Mean   :122          Class :character  Mode :character
mode    : 0.0000   mode    : 0.0000   mode    : 0.0000   mode    : 0.0000   mode    : 0.0000
num-of-doors    body-style      drive-wheels    engine-location  wheel-base
Min.   : 86.60    Min.   :141.1        Length:205   Length:205   Length:205
1st Qu.:166.3    1st Qu.:64.10       Class :character  Mode :character
Median :173.2    Median :65.50       Mode :character  Mode :character
Mean   :174.0    Mean   :65.91       Class :character  Mode :character
mode    : 0.0000   mode    : 0.0000   mode    : 0.0000   mode    : 0.0000   mode    : 0.0000
width      height      curb-weight      engine-type
Min.   :47.80    Min.   :1488        Length:205   Length:205
1st Qu.:52.00    1st Qu.:2145       Class :character  Mode :character
Median :54.10    Median :2414       Mode :character  Mode :character
Mean   :53.72    Mean   :2556       Class :character  Mode :character
mode    : 0.0000   mode    : 0.0000   mode    : 0.0000   mode    : 0.0000   mode    : 0.0000
wheel-base    length      width      height      curb-weight      engine-type
Min.   : 86.60    Min.   :141.1        Length:205   Length:205   Length:205
1st Qu.:166.3    1st Qu.:64.10       Class :character  Mode :character
Median :173.2    Median :65.50       Mode :character  Mode :character
Mean   :174.0    Mean   :65.91       Class :character  Mode :character
mode    : 0.0000   mode    : 0.0000   mode    : 0.0000   mode    : 0.0000   mode    : 0.0000
num-of-cylinders engine-size      fuel-system      bore      stroke
Min.   : 2.00    Min.   : 61.0        Length:205   Length:205   Length:205
1st Qu.: 4.00    1st Qu.: 97.0       Class :character  Mode :character
Median : 4.00    Median :120.0       Mode :character  Mode :character
Mean   : 4.38    Mean   :126.9       Class :character  Mode :character
mode    : 0.0000   mode    : 0.0000   mode    : 0.0000   mode    : 0.0000   mode    : 0.0000
compression-ratio horsepower      peak-rpm      city-mpg      highway-mpg
Min.   : 7.00    Min.   : 48.0        Length:205   Length:205   Length:205
1st Qu.: 8.60    1st Qu.: 70.0       Class :character  Mode :character
Median : 9.00    Median : 95.0       Mode :character  Mode :character
Mean   :10.14    Mean   :104.3       Class :character  Mode :character
mode    : 0.0000   mode    : 0.0000   mode    : 0.0000   mode    : 0.0000   mode    : 0.0000
horsepower      peak-rpm      city-mpg      highway-mpg      price
Min.   : 4150    Min.   :13.00        Length:205   Length:205   Length:205
1st Qu.:4800    1st Qu.:19.00       Class :character  Mode :character
Median :5200    Median :24.00       Mode :character  Mode :character
Mean   :5125    Mean   :25.22       Class :character  Mode :character
mode    : 0.0000   mode    : 0.0000   mode    : 0.0000   mode    : 0.0000   mode    : 0.0000
city-mpg      highway-mpg      price
Min.   :16.00    Min.   : 5118        Length:205   Length:205   Length:205
1st Qu.:25.00    1st Qu.: 7775       Class :character  Mode :character
Median :30.00    Median :10295       Mode :character  Mode :character
Mean   :30.75    Mean   :13207       Class :character  Mode :character
mode    : 0.0000   mode    : 0.0000   mode    : 0.0000   mode    : 0.0000   mode    : 0.0000

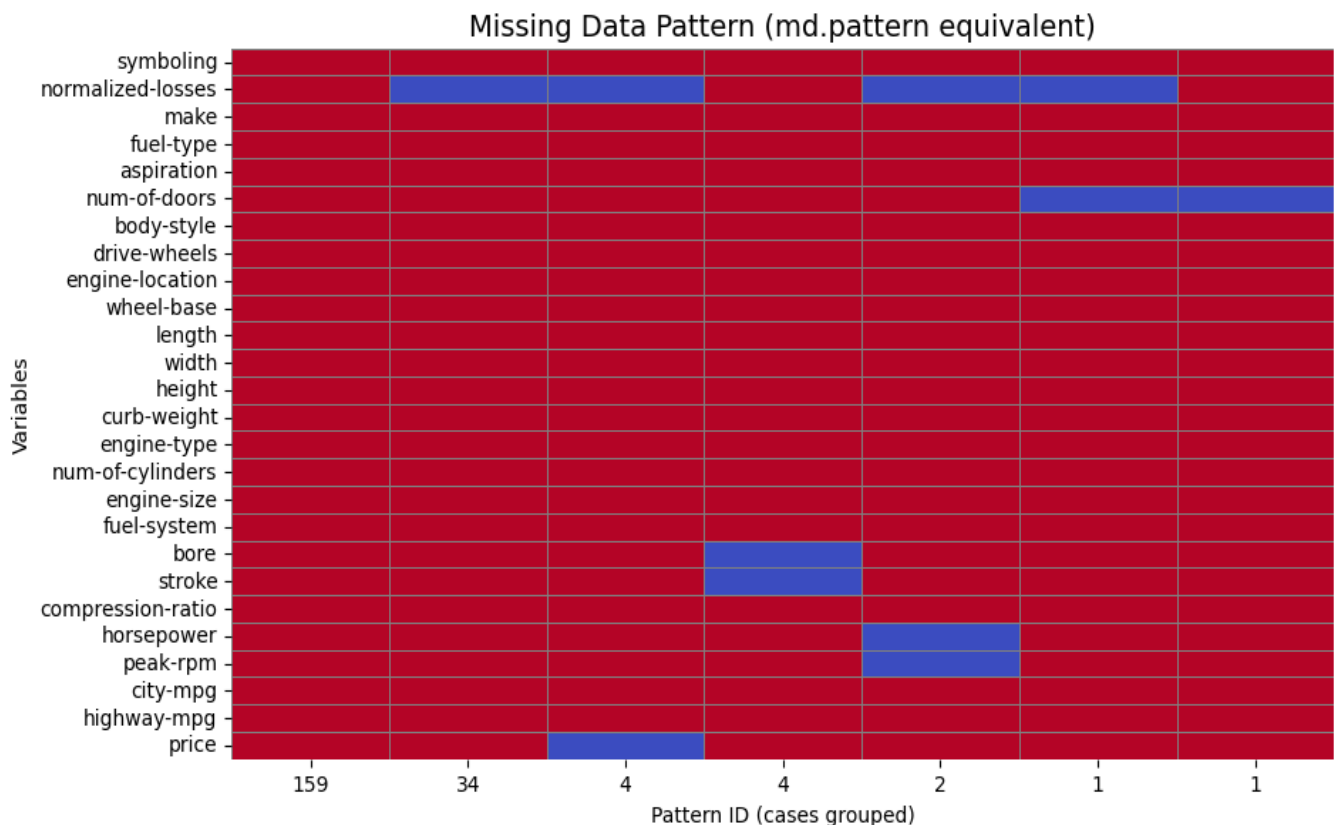
```

Step 4: Checking Missing data pattern:

R Code: `t(md.pattern(b1))`

But we are doing this pattern in python.

Output: The output will represent if there are any patterns within the missing dataset.



Step 5: Delete those rows/tuples with missing values in target/dependent variable:

We have to delete those tuples which have missing values in the target variable. Otherwise, these will impact our model. Our model should not consider those values.

R Code: `which(is.na(b1$price))` #To identify the tuples

`b2<-b1[-c(10,45,46,130),]` #To Delete those tuples

`View(b2)` #To view the modified dataset

Step 6: To identify which variable have how many missing values:

R Code: `colSums(is.na(b2))`

The output will show that how many missing values are present in the remaining variables in the dataset.

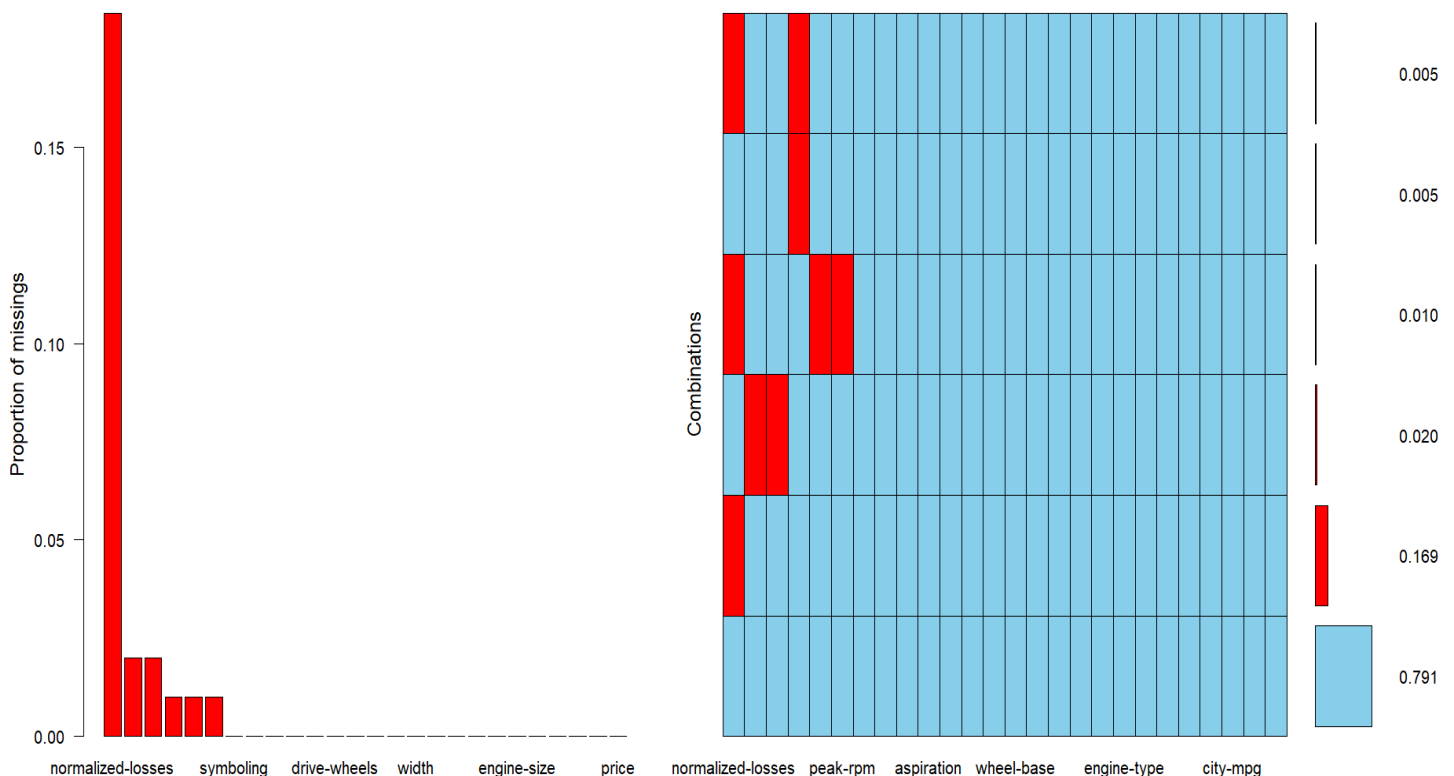
```
> colSums(is.na(b2))
      symboling normalized-losses      make      fuel-type      aspiration      num-of-doors
      0              37              0              0              0              2
body-style      drive-wheels      engine-location      wheel-base      length      width
      0              0              0              0              0              0
      height      curb-weight      engine-type      num-of-cylinders      engine-size      fuel-system
      0              0              0              0              0              0
      bore      stroke      compression-ratio      horsepower      peak-rpm      city-mpg
      4              4              0              2              2              0
highway-mpg      price
      0              0
```

Clearly it is visible, Normalized losses have so many missing values compare to other variables. Also, Bore, Stroke compression-ratio, horsepower and peak-rpm has a few missing values. We have to analyze the missing trend of these values and impute them before building our model.

Step 7: Using `aggr()` plot to visualize the pattern and amount of missing data in the dataset.

R Code: `aggr(b2,pos=1,numbers=T,las=1,sortVars=T, labels=names(b2))`

Output: The missingness of the dataset and the pattern of the missingness in the variables will be visible.



Step 8: Saving the modified dataset in the name of "Import_New".

R Code: `write.csv(b2, "C:/Users/Biswaditya Saha/OneDrive/Desktop/XBS-MBA-BA-3rd SEM/Multi Variate Analysis-Shuvendu Sir/R Studio/Import_New.csv", row.names = FALSE)`

The New Dataset will be saved in the local folder.

Step 9: Checking the type and pattern of the missing value using Parallel box plots to identify the imputation method.

Python Code: independent_vars = ['normalized-losses', 'num-of-doors', 'bore', 'stroke', 'compression-ratio', 'horsepower', 'peak-rpm']

for col in independent_vars:

missing_col_name = f'{col}_missing'

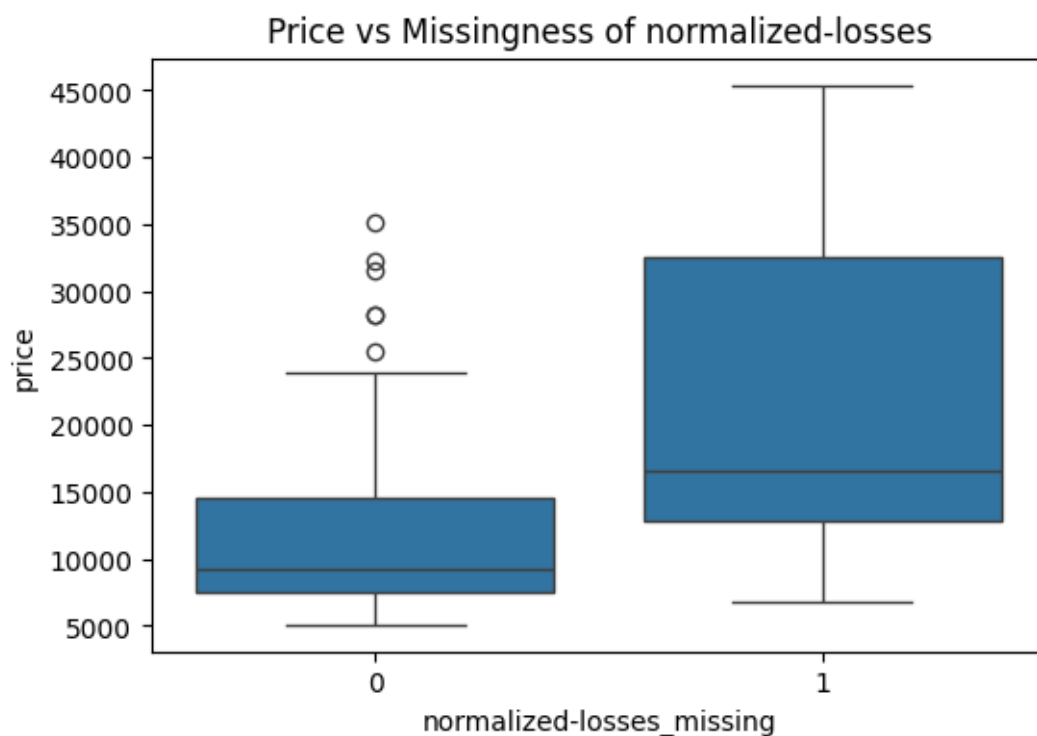
plt.figure(figsize=(6, 4))

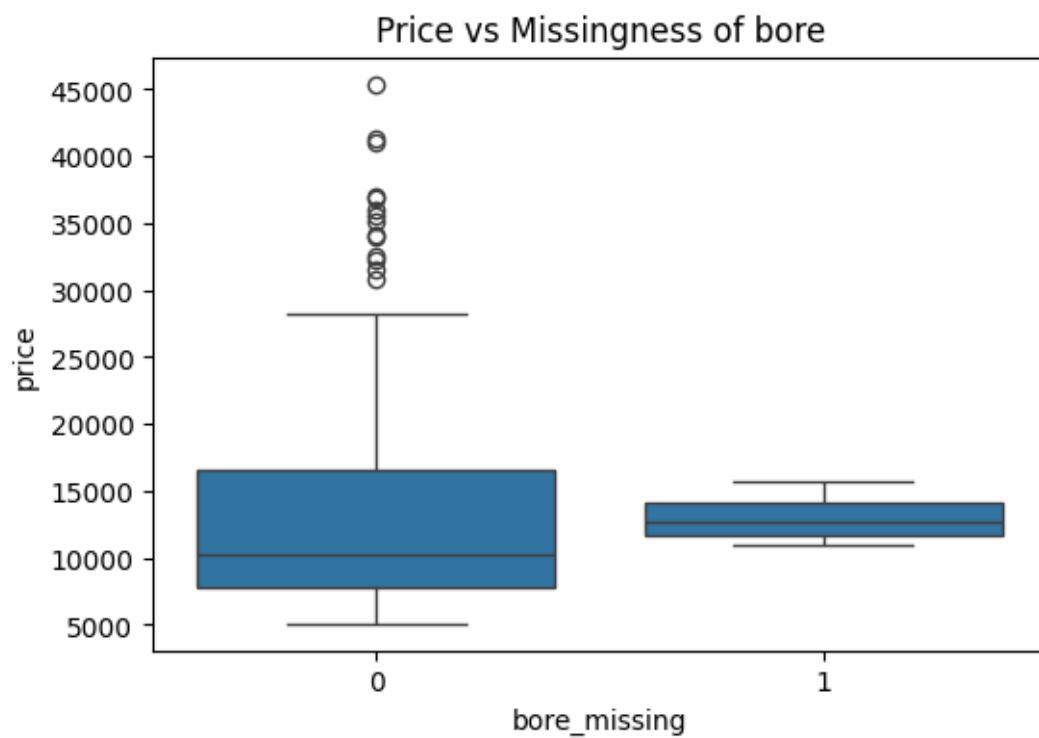
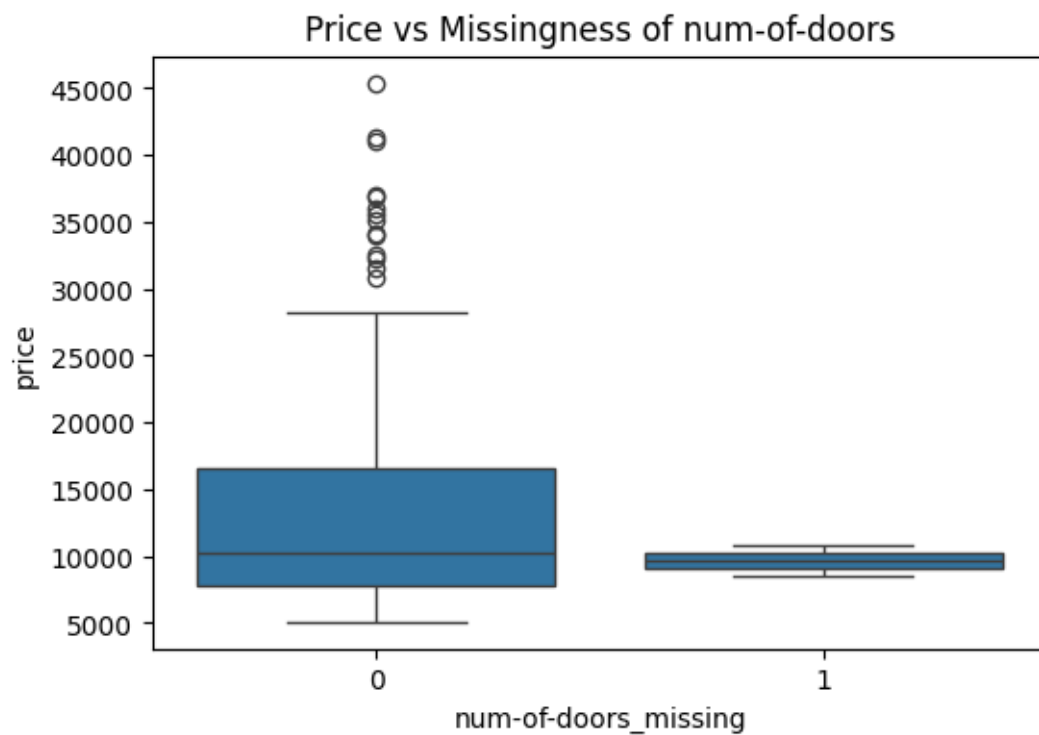
sns.boxplot(x=missing_col_name, y='price', data=df)

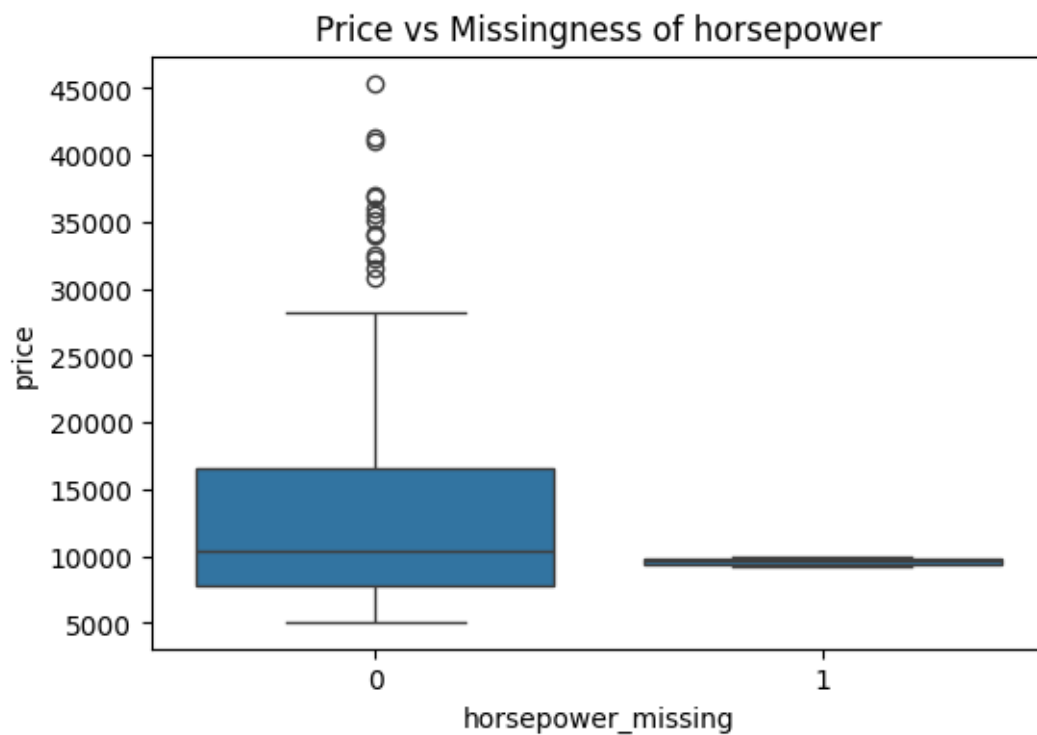
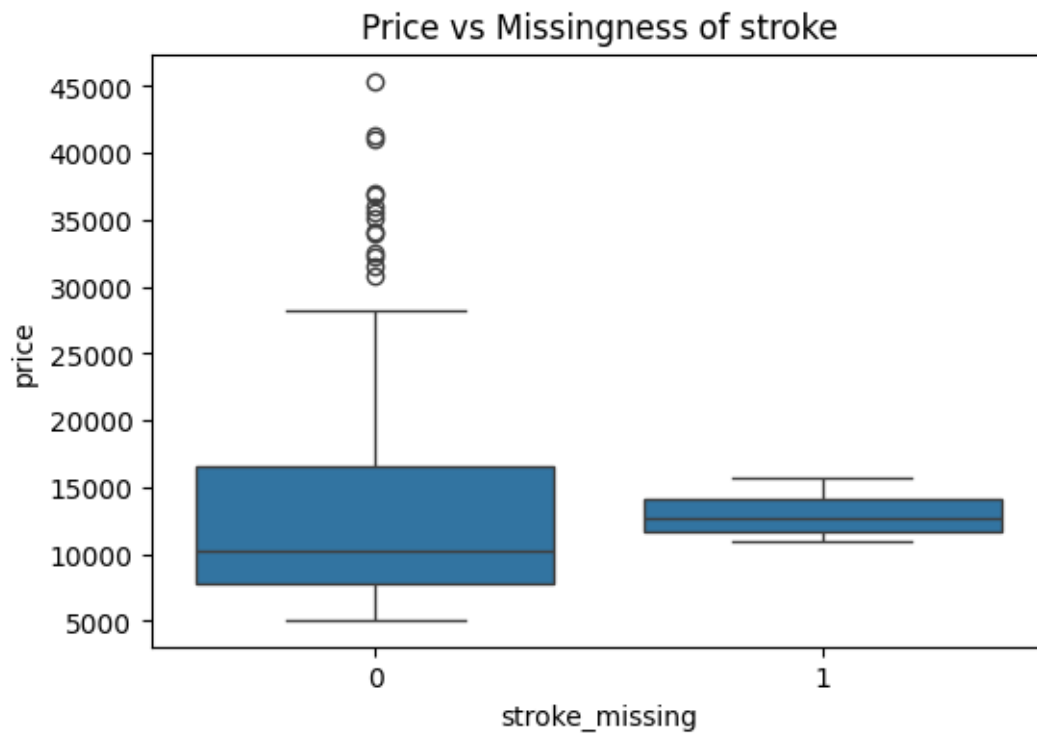
plt.title(f'Price vs Missingness of {col}')

plt.show()

Output: The output will show the missingness of the variables with the price values. This will help us to choose the imputation method of the missing values in the independent variables.







Only these five independent variables have missing values in the dataset. Except Normalization losses, all the other missing data are MCAR type. So, we can choose the central tendency imputation method to impute those values. Also, for Normalized losses, we need to check the dataset further.

Step 10: Imputation of no. of doors:

For dodge makers, sedan types of cars the no. of doors is always four.

For Mazda makers, sedan types of cars the no. of doors is always four.

```
R Code: which(is.na(b2$num-of-doors`))
```

```
b2[27,6]<-"four"
```

```
b2[61,6]<-"four"
```

```
View(b2)
```

Step 11: Imputation of Bore:

As the value is numeric and the data is almost evenly distributed so we can use the mean imputation method here.

```
R Code: b2$bore[which(is.na(b2$bore))]=mean(b2$bore,na.rm = T)
```

```
View(b2)
```

Step 12: Imputation of Horsepower & Peak RPM:

Both the data missing is in Renault Maker car data, and the whole data set have only those two Cars of Renault Maker. Deleting the Renault Car maker data will not impact much as in both the cases the data is missing.

```
R Code: which(is.na(b2$horsepower))
```

```
b2<-b2[-c(127,128),]
```

```
View(b2)
```

Step 13: Imputation of Normalized Losses:

In the dataset, the missingness of the “Normalized Losses” are almost normally distributed. So, we can impute the missing values with mean imputation method.

```
R Code: boxplot(b2$`normalized-losses`)    #To Check the data distribution pattern
```

```
hist(b2$`normalized-losses`)    #To Check the data distribution pattern
```

```
b2$`normalized-losses`[which(is.na(b2$`normalized-losses`))]=mean(b2$`normalized-losses`,na.rm = T)
```

```
View(b2)
```

Now the dataset is completed. I have saved the dataset in name of “Import_New”.

Now based on this preprocessed and cleaned dataset we have to run the linear regression model.

Step 14: Checking the correlation heatmap matrix to identify the relationship between the variables.

Python Code:

```
import pandas as pd

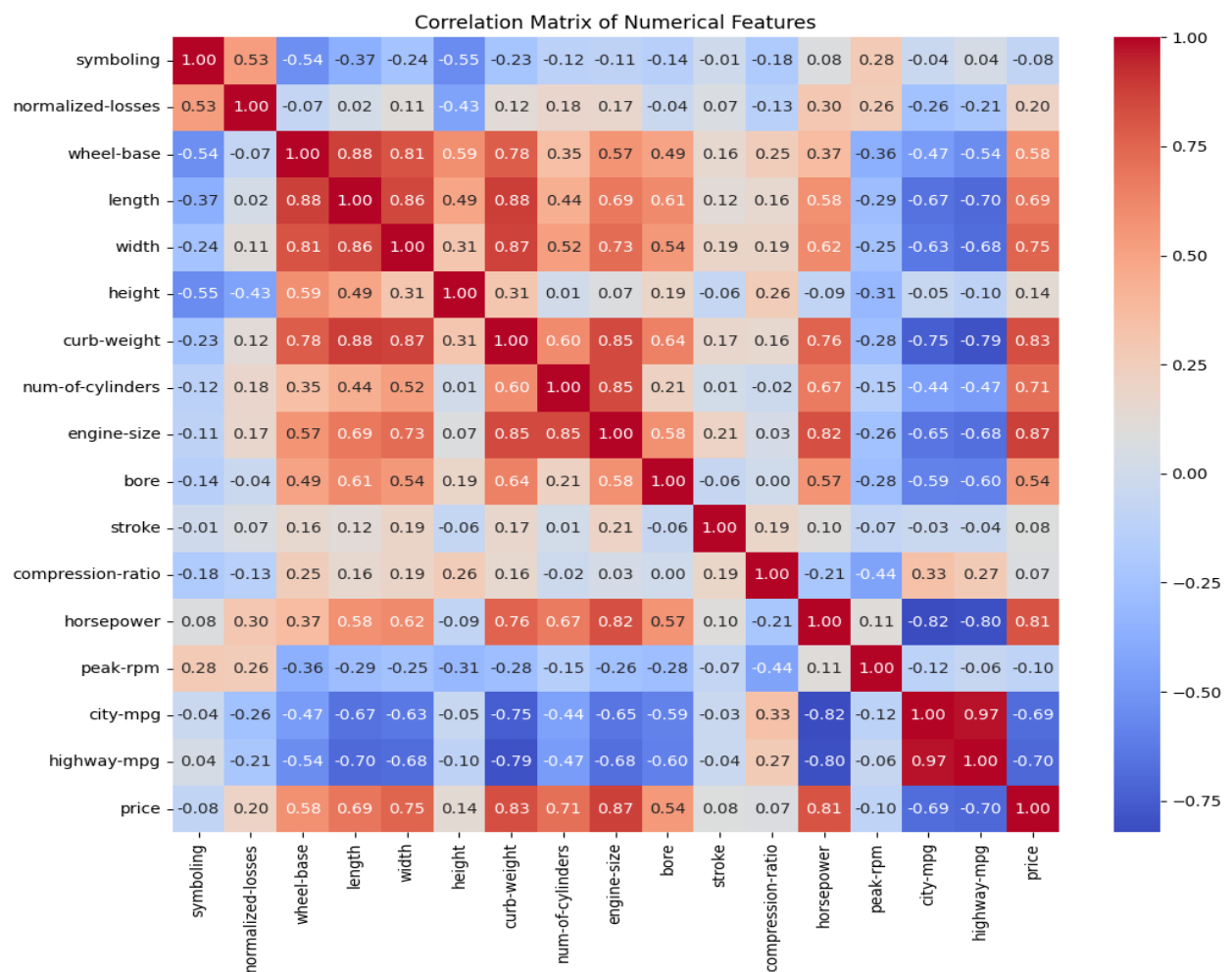
df = pd.read_csv('/content/Import_New.csv')
display(df.head())
display(df.info())

numeric_df = df.select_dtypes(include=['number'])
correlation_matrix = numeric_df.corr()
display(correlation_matrix)

import seaborn as sns
import matplotlib.pyplot as plt

plt.figure(figsize=(12, 10))
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm', fmt=".2f")
plt.title('Correlation Matrix of Numerical Features')
plt.show()
```

Output:



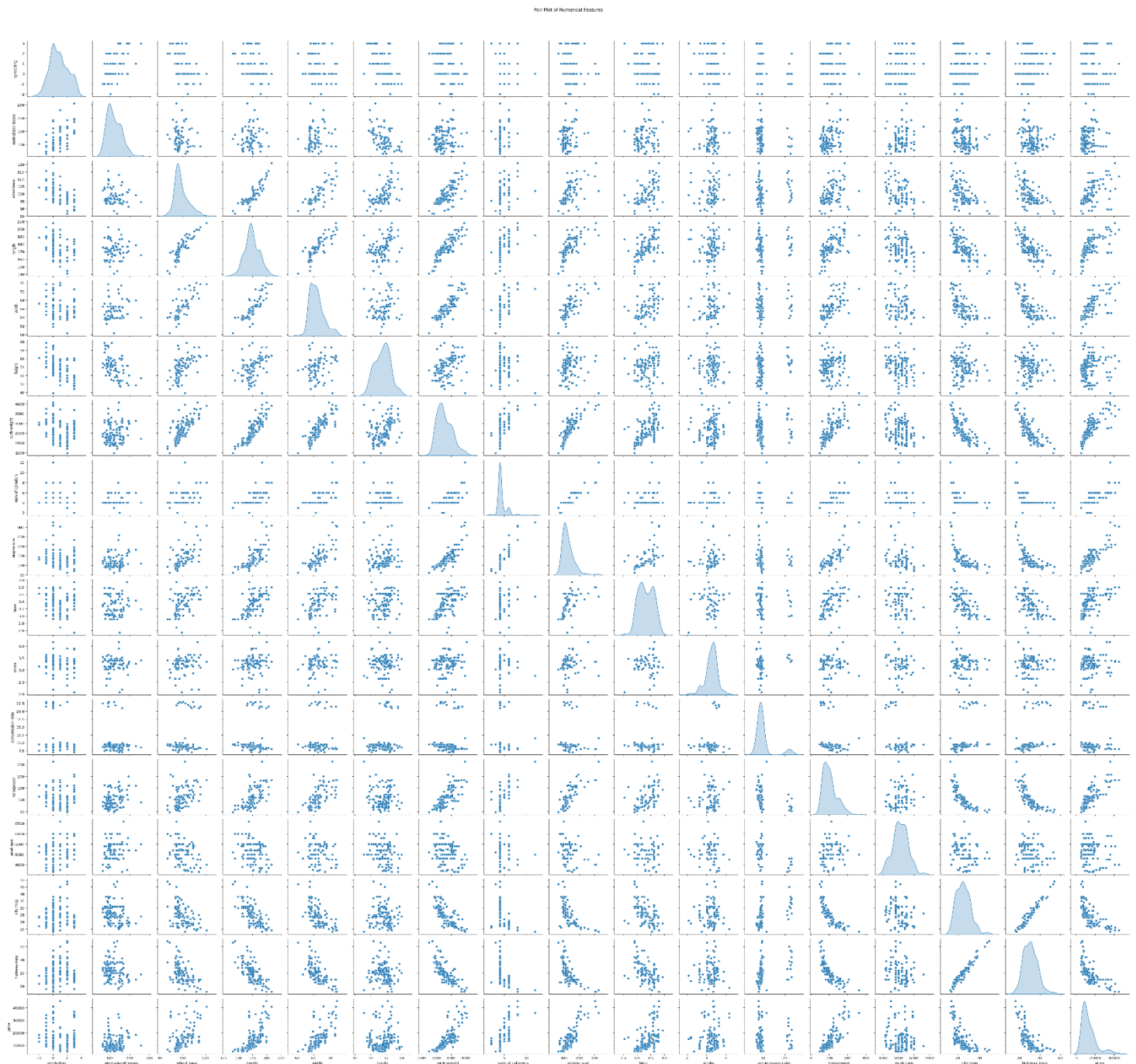
From the Correlation Matrix it is clear that the target variable has high correlation with: Length, Width, curb-weight, num-of-cylinders, engine size, horsepower, highway-mpg, city-mpg.

Step 15: Checking the pair plots:

Python Code: `sns.pairplot(numeric_df, diag_kind='kde')`

`plt.suptitle('Pair Plot of Numerical Features', y=1.02)`

`plt.show()`



Step 16: Dividing the dataset into train and test dataset:

R Code: #Split the dataset into train & test data

```
set.seed(231)
data=sample(2,nrow(b2),replace=T, prob = c(0.8,0.2))
data
train1=b2[data==1,]
test1=b2[data==2,]
train1
test1
```

Step 17: Final Model Building:

Let's just consider the variables which have high correlation with Price. They are:

Length, Width, curb-weight, num-of-cylinders, engine size, horsepower, highway-mpg, city-mpg.

Now, Length and Width have strong correlation between them. So, we can take any of them in the model.

No. of cylinder doesn't have high impact, Also, highway_mpg and city_mpg.

Final Model will Consist: Engine Size, Horsepower, Width/Length.

R Code: `model_Price <- lm(price ~ width + horsepower + `engine-size`, data = train1)`
`summary(model_Price)`

```
> #Final Model
> model_Price <- lm(price ~ width + horsepower + `engine-size`, data = train1)
> summary(model_Price)
```

Call:

```
lm(formula = price ~ width + horsepower + `engine-size`, data = train1)
```

Residuals:

Min	1Q	Median	3Q	Max
-8613.1	-1882.6	52.3	1578.3	14284.7

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	-68126.58	11420.18	-5.965	1.53e-08	***
width	976.14	186.13	5.244	4.93e-07	***
horsepower	55.66	12.47	4.464	1.52e-05	***
`engine-size`	89.70	12.36	7.257	1.65e-11	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3497 on 159 degrees of freedom

Multiple R-squared: 0.8254, Adjusted R-squared: 0.8221

F-statistic: 250.6 on 3 and 159 DF, p-value: < 2.2e-16

From the output we can check that,

Dependent Variable: Price

Independent Variable: Width, Horse Power, Engine Size

Residuals:	Min	1Q	Median	3Q	Max
	-8613.1	-1882.6	52.3	1578.3	14284.7

Multiple R-squared: 0.8254

Adjusted R-squared: 0.8221

Model explains 82% of price variation.

Step 18: Check on the Test1 Dataset:

MSE Value	RMSE Value	MAE Value	R ² Test Value
10333605	3214.592	2160.373	0.717997

```
R Code: predictions <- predict(model_Price, newdata = test1)
```

```
mse <- mean((test1$price - predictions)^2)
```

```
mse
```

```
rmse <- sqrt(mse)
```

```
rmse
```

```
mae <- mean(abs(test1$price - predictions))
```

```
mae
```

```
SSE <- sum((test1$price - predictions)^2)
```

```
SST <- sum((test1$price - mean(test1$price))^2)
```

```
R2_test <- 1 - SSE/SST
```

```
R2_test
```

The Model can predict 72% values of the test dataset accurately.

Step: 19: Visualize Actual Datapoints vs Predicted Datapoints:

